FRED TALKS

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Ownership

THORSTEN BALL · 08 JUL 2026 · SOURCE

Thoughts on ownership I sent to the Amp team in internal note

The following is an internal Slack message I sent to my Amp teammates after I had a conversation with one of them about ownership. It's only lightly edited.

I shared it before, but not here, because I didn't think too much of it. Then today, someone said their CTO shared my post with them and I thought: well, now I have to put it in the newsletter, don't I? So here we are.

Below, after the Slack post, I added some thoughts on juniors on how I see this advice applying to them.



Just had a (great) conversation about ownership and engineering here and I realized that I often use the phrase "ownership" or allude to it, but haven't explained what "ownership" means to me in a while.

So, ownership.

If I ask you "can you own this?" or "can you take care of this?" or "are you on it?" — what I'm doing is I'm asking you to *own* it, to own the solution of a problem from end to end. From "we have a problem" to "we don't have to think about it again."

That means, when you say that you're owning something, the expectation is that you…

- Think about what the problem actually is. Maybe you already have a solution in mind, without having thought about what we're actually trying to solve here. Maybe you think "the problem is that we need to migrate from using X to using Y", but that's not a problem, that's a solution. The problem is likely something like "performance is bad", "it's not stable", "it fails for customer x". Maybe there's other possible solutions to that? Think about those. What are the tradeoffs? What's the best solution to go with considering these tradeoffs?

- Think about edge cases. What are they? Which ones are important? Which ones can we ignore?

Look at the past history of this blog. There are many blog posts abou

So mine is a trick. People feel more and more programming is complete

This is why yesterday, on X, I said that I believe many programmers a

1. You can now generate a lot of code, even *not* accounting for the

2. LLMs are very good at writing locally optimal code, and are worse

3. The working day is 8 hours. If you read the code, it is a tradeoff

Controlling the ideas. Do you remember this phrasing from the Mythica

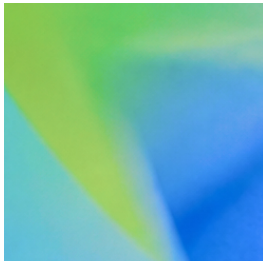
Matteo Collina yesterday asked me, in reply to my tweet: but didn't y

Fable and GPT 5.6 reviews to the sorted sets memory saving are going

:

Keep reading

View all



Research acceleration: The view inside OpenAI

ResearchSep 6, 2026

[VIDEO OMITTED]

GPT-6 Astra: A new generation of intelligence

ResearchSep 3, 2026



Safety overview: GPT-6 Astra

SafetySep 3, 2026

Control the ideas, not the code

ANTIREZ.COM · 18 JUL 2026 · SOURCE

- Think about failures. Network failures are a given, for example. How do we handle them? Retry? Well, how often? How long?

- Think about data flow. How much data is involved here? Does data need to be migrated? Cleaned up? How can I get my hands on data to properly test this? What invariants are in the data? What assumptions do I have about the shape of the data that I haven't confirmed yet?

- Think about how you'd test this. How can I know that what I built is correct or not? Are tests enough? Do I need to manually poke at things? Is the difference visible on a screenshot or in a video?

- How would we announce this? How do we communicate it? Can you picture it? How does it fit into the larger picture of our roadmap? Questions or concerns in that area — push back! ask!

- Do the work, with precision, with care, with a sense of urgency, with calmness. Do not half-ass things. Before you merge, ask yourself: am I proud of this? would I show this to John Carmack and say "here's what I built, under these constraints, with these tradeoffs?"

- Test it manually. Yes, there's automated tests. But in 99% of cases you can manually test or confirm that what you built works: you can run it yourself, you can ask an agent to run through test scenarios, you can poke at the data before and after, you can take screenshots, you can make a demo. Are you sure that what you did actually solves the problem?

- Make sure it lands in production and *works in production*. Is it deployed? Did the deploy fail? Do you need to activate a feature flag? Does the feature flag work? Can you use it in production? Can you confirm it's actually deployed?

- If you think your colleagues needs to know about this change, because it's new feature they should all test, or it's a new convention in codebase, or maybe it's a tricky thing everybody needs to be aware of, or something else: let them know! Do not underestimate peripheral vision: knowing that person X yesterday changed the behavior of how Z works might save person Y three hours of debugging today when a bug report related to Z comes in.

- Do customers need to know? Who reported the bug? Who's blocked? Let them know.

- Does the world need to know? Announce that it's out.

- Are there follow-ups? Do you need to check on what you shipped in the logs? A week later maybe?

Yes, that's a lot. And there's actually more, because I'm sure I forgot some stuff.

But that's how you build a product in a small team. We don't have PMs, we don't have a Q&A department. We're small, but we're *great*, we can do all of that.

And it's always okay to ask for help, it's okay to ask questions, it's okay to redo things and triple-check. What's not okay is to implicitly assume that someone else will do the things here that you haven't thought about.



"How does this apply juniors? You can't expect them to really do all of that?"

I've been asked these questions, or variations of them, after I shared the thoughts above and here's my answer.

I do *not* expect juniors to *do* all of these things right away. But I would expect them to read through the list and *aspire* to one day be able to do all of these things. Until then, they can and should ask for help.

In fact, I don't expect *anybody* to *always* do *all of these things* for *everything*. It's a mental checklist of things to consider — problem, edge cases, tradeoffs, deployment, customers, messaging, … — but for quite a few things there aren't edge cases to consider. Or big tradeoffs to weigh. Or deployment is a solved problem. And maybe someone else does the messaging for you.

And I imagine that most of these things you shouldn't even consider when you work in a company with, say, 5000 employees. I've never worked in a company that large, only startups, so I can't speak to how to successfully ship a software feature at Apple, end to end.

But when you work in a small company in which there's only a single department, when you want to build things you're proud of, when you work with me and you say own something, I expect you to keep these things in mind and run through them before you declare something as done.



You know what *you* should own? A subscription to this newsletter:

As great as the long-term promise of AI may be, the majority of our focus should be on the next few years. We are facing a transition to a world with incredibly intelligent machines, and we need to ensure that transition works out well for humanity. We need to find ways to preserve human agency and enshrine an intrinsic value to being human, in a world where most tasks could be performed by AI. To prevent extreme concentration of power in a world where undertakings that would have taken thousands of experts now will be achievable by a few people operating a large computer. And to ensure that humans remain in control of the future and are not left behind by unchecked progress, brought about by an alien intellect exceeding our own.

Currently I believe that no lab has solved alignment and monitoring to a sufficient degree to continue responsibly scaling at maximum speed for much longer. I expect and hope for voluntary slowdowns to become commonplace until shared safety bars are established. And I believe that international coordination on future AI development needs to become a top priority for governments around the world.

- Alignment
- 2026

Author

Jakub Pachocki

Footnotes

1. This included scaling self-play(opens in a new window), robotics(opens in a new window), and, in retrospect most notably, scaling recurrent networks to model language(opens in a new window), which was a precursor to the GPT line of research.

   [find in text]

2. A secondary reason for this design was preventing distillation. However, maintaining CoT monitorability has explicitly been the bigger priority for us throughout development.

   [find in text]

The concrete bits of progress we've made on alignment and monitoring have generally been very intertwined with general AI progress. Great examples are RL from human feedback, which was key to training early AI assistants, and the aforementioned chain-of-thought monitoring, which was enabled by advances on reasoning models. We must focus the increasingly automated research process on developing new such insights, algorithms and theories, and iteratively build up safety cases for more capable AIs.

Scaling AI systems has to be constrained by our confidence in safety. We need to evolve commitments like the Preparedness Framework or Responsible Scaling Policy into widely mandated safety bars for continued development. These can be enforced by a network of third-party auditors, by government agencies or by international bodies.

The core challenge of automating AI research is not "getting there" - it is getting there in a way that keeps people a part of the continued improvement process, and leaves the future in humanity's hands.

What is next?

As we outlined recently with Sam, OpenAI prioritizes work in service of three north stars:

1. Navigating the next period of AI progress, by building an automated AI researcher, iterating with it on the alignment problem and finding ways for people to remain part of the self-improvement loop.

2. Delivering the benefits of scientific progress and economic growth that very intelligent machines enable.

3. Empowering everyone individually with a personal AGI.

I have focused in this essay only on the first point, as I believe it is by far the most urgent. However, I hold a deep hope and appreciation for the benefits that further technological progress will bring. Future aligned AI could advance science, develop new therapies, and bring about broad material abundance. Friendly and honest AI can help people navigate difficulties they face in their life and meaningfully improve their happiness and sense of fulfillment. OpenAI puts a tremendous amount of effort into bringing these benefits about. One current example I am proud of - and my loved ones have found helpful - is the deep investment into ChatGPT's ability to provide health information.

The lies I used to tell myself

CATE HALL · 23 JAN 2026 · SOURCE



1. What doesn't kill you makes you stronger.

When I was 31, my husband of four years, who I loved more than my own life, left me completely out of the blue. We'd had an incredible, epic romance — drawn together like magnets, we married on our third date and spent the years that followed burrowing ever closer together. Maybe that was the problem — I can't say for sure, because I never got a satisfying explanation out of him. One day I was the protagonist of a Princess Bride-worthy love story; 48 hours later, he moved out.

From the time we met, I had never contemplated the possibility of living without him, and I didn't want to. I wanted to die for a long time — at least a couple of years. But as I slowly learned to sequester away thoughts of him and the emotions they brought up, I became convinced of a silver lining: The experience would make me impenetrable. If I survived it, I would know I could survive anything, and nothing would be able to hurt me again.

It took the better part of a decade and another deeply identity-shifting relationship, with the man who became my second husband, for me to see that I had it all wrong: Impenetrability means avoiding the parts of life that are most life-like. There's a reason the visual metaphor for love is an arrow.

The near-fatal catastrophes of our lives don't always make us stronger. Surviving one hard thing does provide evidence that you can survive other hard things, which is easy to *mistake* for getting stronger. But in reality they often just make us brittle, make us dissociate and flee from the situations that remind us we can be broken.

2. When you know, you know.

Mario Gabriele of the Generalist had a great "50 Things" style post a few months ago that I found myself nodding along to almost the whole way through. The glaring exception was Number 33: "Love is easy. The difference between bad relationships and meeting your spouse is not a matter of degree, but category. Compatibility is unignorable and effortless."

I believed this completely, once. I've since come to believe that what's unignorable and effortless is chemistry. Chemistry can be powerful enough to turn your life sideways, but it's not the same thing as compatibility. Compatibility is litigated on the timescale of years and decades; it's not a momentary feeling but a description of what happens during an extended course of change. Do you grow toward one another, or do you grow apart? Does the relationship help you move closer to the person you aspire to be?

I "just knew" with my first husband. I wasn't sure with my second. Four years in, the signs have flipped.

3. If it were a good idea, someone would be doing it already.

There's a belief, especially common among wonky types, that you don't find $20 bills on the sidewalk — if an opportunity were real, someone would have seized it already. This sounds worldly and wise, but it's actually a form of defensive thinking. It assumes the current way is best, that everything worth doing is being done.

In my experience, this is just false. When I started playing poker, I discovered that physical tells were basically a wide-open field — despite the fact that most pros believed the topic was played out. If you think you see an opportunity others have missed, you should have a story for why they're missing it, and you should ask around to stress-test that story. But if no one can give you a satisfying reason, don't assume one exists. Sometimes the answer really is just inertia or fear.

The risks associated with AI are unfortunately going to grow from here. A very capable agent explicitly trained and instructed to carry out nefarious acts presents a new kind of danger; it is likely to cross the scope of its operator's intent, generalizing into potentially more extremely malicious behavior. The boundary between misuse and autonomous misaligned actions will blur as AI gains more agency. We may be used to thinking of AI as tools, but some agents will be pursuing their own objectives. They will find ways to collaborate with people, by bargaining with, tricking or blackmailing them.

In addition, there are the risks that come from new technologies potentially enabled by AI, such as engineered pathogens.

We will need powerful, aligned AI for defense; to secure infrastructure, to protect against rogue agents in real time, and to invent entirely new protective measures. This will be a primary focus of OpenAI's deployment efforts.

At the same time, even with the uncertainty that comes from anticipated broad AI progress and the need to build defensive systems, we must not let that become an excuse for recklessness. The idea of racing forward at all costs seems absurd once one internalizes the seriousness of the stakes.

Pacing RSI

Machine intelligence playing a larger and larger role in its own development process is a natural conclusion of sustained technological progress. If AI progress continues, machine recursive self-improvement (RSI) will be at the very core of future scientific discovery.

Automated AI research is a more dramatic form of scaling intelligence with compute; and of course as a part of it, AI will improve the computational substrate itself. And similarly to scaling, we focus OpenAI research towards RSI as we believe it is the only way to remain at the frontier of AI research moving forward.

I want to stress that the above words don't imply I think greatly accelerating deep learning research, especially in the short term, is the right collective action we should take as the research community. However, I do think this is where the current path leads, and we all need to make a conscious choice on how to proceed. The main levers we have are either steering the process to strengthen alignment and monitoring alongside the AI and find ways to keep people in the loop; or coordinating to slow down future development as needed to build confidence in these measures.

The best way forward I see currently is a combination of both.

In development since, we have strived to maintain the rule of not supervising the reasoning process. CoT monitoring became an extremely important tool for us in studying how our models generalize from their training distribution, allowing us to observe and analyze not only their actions but also their internal process.

This tool continues to be critical as we study the Astra class of models. However, unfortunately our evaluations indicate our ability to rely on CoT monitoring is progressively diminishing. This comes from a combination of factors.

- Modern reasoning models are used in more complex environments than o1-preview; their reasoning process is increasingly blended with communicating with people, other AIs, and using tools. Many of those interactions have to be supervised, thus blurring the boundary we aim to preserve.
- The AI is becoming better at reasoning about and manipulating its own reasoning process.
- With improved pretraining performance, we also see the models become much smarter even without using verbalized reasoning at all.

These challenges are not necessarily insurmountable. I am hopeful we can develop interventions to improve chain-of-thought monitorability of our models, e.g. by forming a better understanding of the interplay of different optimization objectives and forms of test-time compute the model uses. I also believe there can be great value in combining ideas from CoT and activation monitoring - scaling training of monitors with direct access to network internals, e.g. confessions. We are actively pursuing these ideas. Still, I expect general AI progress to increasingly be bottlenecked by confidence in monitoring.

Scalable defense

The strongest argument I see for continuing to train much smarter models quickly is the need to build defensive systems against the dangers posed by other AI.

A clear risk discussed throughout this year is to cybersecurity: the models are becoming superhuman in their ability to break in and out of computer systems. This expands the scope of risks associated with AI tremendously: agents are going to be able to access any but the most secure infrastructure, and affect a lot of the world directly, even without a physical body. We are currently in a narrow window to use the best available models to significantly tighten security of critical systems.

4. If it's worth doing, it's worth doing well.

Almost nothing is worth doing "well." Many things can and should be done to a minimal standard of quality, because anything more is a waste of effort that can be better allocated. The exceptions aren't things that should be done "well," but things that should be done to an exceptionally high standard, probably a much higher one than you see modeled by the people around you.

This is not semantics; most people really "do everything the way they do anything" — whether that's in a low-effort or high-effort way. But the point of phoning it in on a lot of stuff is to save your energy to expend on the 1 in 1000 things that really matter … and then to actually spend it.

A recent example: I decided that I want to narrate my own audiobook for *You Can Just Do Things*. When I told a friend in the industry this and asked whether they knew any good coaches to work with, they said that hiring a coach wasn't normal or necessary, because my publisher would decide based on a sample whether I was good enough at reading to do it, and if so give me a producer to work with day-of. But it "1 in 1000" matters to me whether the audiobook is so good no one would ever think of returning it on account of the narration, so I'm not interested in the normal-shaped solution that constitutes doing it "well," I'm interested in the one that causes me to come to mind when someone asks this question. In this case, that means hiring multiple coaches and practicing for months.

5. If you like your job, you're doing something wrong.

Back when I was more alienated from my emotions, I thought about work like this: You have a budget of energy to spend, and the goal is to efficiently convert that energy into impact. You identify the activity that generates the highest impact per unit of energy, and then you spam that button for as many waking hours as you can stomach. Is it any wonder I was always burning out?

What I failed to understand is that energy is not fixed. It's dramatically affected by what you choose to do — some activities increase your energy over time, others deplete it. This feels so obvious to me now that I feel kind of stupid including it here, but I know many people (they are endemic to the Bay Area) who still labor under the false belief. Whether you enjoy what you're doing affects your ability to do it year in and year out, and dismissing this as "selfish" doesn't make it untrue. As my husband puts it, the best use of effortful motivation is to engineer your life to require less of it.

6. “All people are insane. They will do anything at any time, and God help anybody who looks for reasons.”

This line, from Kurt Vonnegut, used to neatly encapsulate the way I understood other people — which is to say, not at all. Trying to figure out why people did what they did, or imagine what they’d do next, seemed like a fool’s errand. I was judgmental about it, too: I assumed this was a problem, not with me, but with everyone else.

I didn’t fundamentally move beyond this belief until recent years, when I came into contact with the Enneagram. I imagine there are other personality typing systems that can do something similar, but the Enneagram was magical for me because it clued me into the fact that people can have very different motivational systems from my own that are still internally coherent and produce good things in the world.

People’s behavior was previously incomprehensible to me because I was imputing my own motivational system to them, trying to figure out what would cause me to act the way they did — essentially, concluding that *their* actions made no sense in light of *my* goals. But people have wildly diverse emotional hardware, which determines what’s rational for them. And my particular personality — independent, moralistic, systematic, challenge-seeking, competitive — is unusual, so I was especially poorly placed to measure others by the yardstick of myself.

This shift in perspective was dramatically good for me and for my relationships with other people. It let me see the ecosystem of psychologies as a kind of miracle — wow, it really does take all kinds — rather than a prompt for cosmic horror. And it gave me confidence that, if I invested in understanding people more deeply, they would eventually cease to be total mysteries, prone to bizarre and unreasonable behavior. As far as I can tell, this is the better part of emotional intelligence.

7. Money can’t buy happiness.

The good version of this advice is: You’ll still have to work at being happy, even if you’re rich. Money in your bank account won’t do it for you, and wealth comes with its own pathologies, like endless status-chasing. But the common gloss — that money stops mattering past some modest threshold — is basically false.

You may have heard there’s research showing money doesn’t buy happiness. That research was explosively popular precisely because it was counterintuitive — it betrayed the commonsense intuition that more money means more freedom, more options, more ability to help people you care about. Turns out it was also riddled with methodological problems. More recent work by Matthew Killingsworth (nominative determinism FTW), using large-scale studies that pinged participants about their happiness at random moments, found steady returns to well-being with

the model’s ability to generalize from the situations it has encountered in training. For example, in the OpenAI-Hugging Face incident, the agents preserved a boundary of not social engineering humans. However, they clearly failed to abstain from other actions that were out of scope and went against the spirit of the values they were taught in other settings.

The second approach seeks to leverage the model’s ability to generalize from pretraining data. This can involve crafting alignment-inducing training datasets, or focusing the model on an ‘aligned’ part of the pretraining distribution, as in, for example, the persona selection model. The weakness of this approach lies in the lack of robustness to further optimization pressure. If you take a model that thinks generally ‘aligned’ thoughts, and subject it to enough training where it’s taught to achieve very hard objectives, it can learn to reason in a motivated way: bending the 'aligned' seeming thoughts as needed to achieve the goal. We likely saw an example of such behavior in recent cybersecurity incidents involving a non-OpenAI model.

We invest heavily along the spectrum of approaches spanned by these directions. We also see meaningful progress - GPT-6 Astra is the first model that benefits from some important advancements we have been working on for a long time, and is significantly better aligned than GPT-5.6 Sol. Still, it is important to acknowledge and understand that much more progress is required as models become more capable; and that progress in generalizable alignment may not sufficiently outstrip progress in general model intelligence.

Monitoring generalization

We do not have a satisfactory theory of generalization, and it seems unlikely that we can develop one soon, at least without the help of more powerful AI. Therefore, at present, our ability to empirically validate our alignment techniques is in practice arguably even more important than the alignment techniques themselves.

OpenAI’s primary bet here has been chain-of-thought monitoring. It is based on an appealingly scalable idea: a lot of the model’s capability comes from a verbalized reasoning process (chain-of-thought). If we scale optimization on the outcomes of that process, but do not supervise the process itself, that chain-of-thought has no direct incentive in training to hide any misaligned ideas or objectives. This does not mean the model will learn to externalize misaligned tendencies that don’t rely on using the chain-of-thought; however, it can allow us to monitor exactly the capability increase from reasoning.

We understood the potential significance of chain-of-thought monitoring at the same time we developed reasoning models. When we shipped o1-preview, we deliberately designed the product to hide the chain of thought, to protect it from supervision pressure in the long term[(2)].

Teaching machines to love

Because machine intelligence comes from a fundamentally different process than human intelligence, we cannot assume it adheres to human principles by default, or generalizes from them in a human-like manner. The core problem in AI research is that of *alignment* - getting the AI to "try to do the right thing" by human standards.

For the purpose of organizing practical research directions, I find it useful to distinguish *goal alignment* and *value alignment*.

Goal alignment is broadly: "does the AI try to accomplish the goal set before it?". This can include things like adherence to an instruction hierarchy, or the ability to communicate and collaborate with people, to attempt to understand their objectives. This set of directions has been extremely practically relevant.

Value alignment is a more intrinsic property of the model. It is the ability to hold and generalize from a high-level set of principles; to act "reasonably" even when given unclear or conflicting objectives, or placed in unfamiliar or adversarial situations. An aligned AI should act with honesty and integrity, and love for humanity.

Of course, the boundary between value and goal alignment can be blurry, and truly caring about goals requires attempting to infer the intent and values underlying them. However, generally when I talk about the long-term importance of alignment research, I am referring to value alignment.

The fundamental challenge of AI alignment is generalization. As machines become smarter, they find themselves working on higher-level concepts, and placed in environments increasingly different from those they encountered in training. They can fail at generalizing from the values taught and reinforced in their training process to those new situations; and it can be hard for us to be sure how they will act. This is made even more difficult by the fact the overall ecosystem the AIs are used in is changing very quickly; for example, AIs trained today need to be robust to interacting with a variety of other AIs. Crucially, we need future AIs to continue to hold human values regardless of whether they believe they're under human supervision.

There are two major classes of currently practically employed methods for alignment training.

The first is encouraging aligned behavior as part of goal-oriented reinforcement learning. Model's actions are evaluated (usually by AI) for being consistent with a given preference model, "spec" or "constitution", and rewarded appropriately. This approach can be very effective in the average case, and is a core part of how modern AI assistants are made. Unfortunately, it can also be brittle and strongly relies on the coverage of training oversight and

rising income, no plateau in sight. The myth persists because it serves a psychological function: It helps people without money feel better about their situation, and helps the wealthy downplay their advantages. But it's not true.

8. Intuition is fake and rationality solves everything.

I used to be horribly judgmental of people who made decisions based on intuition, as in, "I just do what feels right." And then I noticed that whenever I overrode my intuitions, I made *terrible* decisions. I hired the wrong people, started projects with the wrong people, dated the wrong people, ate food that made me feel bad, did forms of exercise I found unfun and injurious. I don't have a satisfying theory for why this is, but I know what happens: When I shift from emotional intuition to a logical story about a decision, it's much easier for my reasoning to get hijacked by plausible-sounding directives that turn out to be nonsense.

It took me a long time to learn this, and that might have something to do with my training in law and poker. Overriding intuition in favor of explicit reasoning makes sense in contexts that reward systematic rationality, where you will get separated from your money if you count on gut feelings. But most of life isn't a closed system with enumerated rules and clear outcomes. If you try to explicitly name all the factors that make someone a good friend or co-founder, you might come up with some good indicators, but your crude map will also mislead you about the territory.

This is why "wisdom is knowledge that can't be transmitted." Wise people have navigational skill, not a long list of rules.

Sign up to be notified when my book, You Can Just Do Things, is available for purchase.

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September 6, 2026

Safety Research

Author: Jakub Pachocki, Chief Scientist at OpenAI

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[AUDIO OMITTED]

Share

In mid-2023, within the “RLSlow” research project, we saw the first results that gave us confidence that we will be able to scale the training of reasoning models, unlocking the capability of pretrained models to form their own chains of thought. Szymon and I spent that night at the office, thinking not about the incredible benchmark numbers, products, or scientific results that this technology will deliver - but rather, trying to process the sobering fact we will actually see machines meaningfully smarter than ourselves in our lifetime, and we already see the shape of these systems; wondering how to alert people to the significance of this.

Three years later, reasoning language models are a rapidly growing part of the economy and starting to push the boundaries of science. They are able to operate computers and graphical interfaces, collaborate with people and each other, and carry out research projects. They are also transforming the landscape of computer security, and in that present clear new dangers.

A lot of new research happened in this period, and our understanding of these systems is again a little different than it was in 2023. Based on internal results, I have a strong expectation that this speed of progress could be sustained into recursive self-improvement. If AI development continues along its current path, the systems we’ll see in the next few years are likely to represent further capability jumps of equal or larger magnitude, and to increasingly drive their own development.

This is a time that calls for extreme caution. I am concerned no one is prepared for the consequences of a continued rapid rise in machine intelligence. OpenAI will continue to seek technical solutions to alignment and monitoring, to build defensive systems and unilaterally withhold further scaling as needed; however, I believe broader interventions are required.

Intellect we don’t fully understand

At a high level, progress in machine intelligence is driven by increasing computational power. We at OpenAI deeply internalized this around 2017, after seeing consistent returns to scaling across multiple research projects(1). As a result, we sought out access to much more compute than we had originally planned, and increasingly oriented our research around a small number of very scalable directions. We believed that was the only way for us to be at the frontier of AI research, and influence the impacts of AGI.

There are new algorithms that have been developed along the way, new feats of ingenuity from teams and individual researchers. I see them largely as discoveries along the path of scaling; the science of deep learning is still nascent, and meaningful algorithmic progress tends to correlate with access to compute. If you zoom out to a multiple-year horizon, AI is continuing to become more intelligent as it is scaled to larger computers.

And, in line with Ray Kurzweil’s predictions from the end of the XXth century, we now find ourselves at the moment in history of computing where machine intelligence is starting to exceed that of humans in transformative ways.

AI is *grown* more than *designed* - it is, to first degree, the product of repeating a straightforward optimization step many times on a hard-to-imagine amount of compute. This results in an incredibly complex system that works through abstract concepts and can simulate facets of human behavior. We can discover various insights about little mechanisms that emerge within this system, in a process similar to neuroscience - and, similarly to neuroscience, its overall action evades a description we can fully understand.

The study of deep learning-based AI is largely an experimental science. We put a lot of effort into building principled algorithms and making testable predictions, but fundamentally, our large-scale training runs are *experiments*, and we are sometimes surprised by their results. Moreover, as the systems become more capable, the results become harder to interpret.

This is made more complicated by the current algorithms generally improving easy-to-measure capabilities faster than those hard to objectively quantify. We spend a lot of time trying to understand how capabilities generalize, and what to prioritize to advance the skills that are going to be most relevant in the next few years. For instance, we believe we could make the models better at specifically mathematics research with additional focus, but we do not prioritize this direction because of the urgency we feel about RSI and automated alignment research, as I will discuss later.

The intelligence produced by scaling deep learning is not directly comparable to human intelligence. To become very relevant in the real world - very useful or very dangerous - the AI does not need to match or exceed all human capabilities; it just needs to surpass enough of them. And as it continues to surpass humans on more and more axes, it is becoming increasingly difficult to understand exactly how capable it is.